

Table 3: Balanced accuracy on Varied Deception for the DYL probe and two follow-up question variants (DYL Honest, DYL Truthful). The original “did you lie?” framing achieves the highest median balanced accuracy, though per-model differences are small and inconsistent.

model	DYL Probe	DYL Honest Probe	DYL Truthful Probe
MiniMaxAI/MiniMax-M2	0.74	0.72	0.72
MiniMaxAI/MiniMax-M2.7	0.84	0.80	0.83
Qwen/Qwen3-14B	0.80	0.77	0.75
Qwen/Qwen3-30B-A3B	0.82	0.75	0.74
Qwen/Qwen3-32B	0.76	0.75	0.74
Qwen/Qwen3-4B	0.68	0.64	0.67
Qwen/Qwen3.5-122B-A10B-FP8	0.92	0.93	0.93
Qwen/Qwen3.5-27B	0.93	0.90	0.90
Qwen/Qwen3.5-35B-A3B-FP8	0.88	0.88	0.87
Qwen/Qwen3.5-4B	0.82	0.85	0.88
Qwen/Qwen3.6-27B	0.88	0.89	0.88
Qwen/Qwen3.6-35B-A3B	0.86	0.86	0.92
XiaomiMiMo/MiMo-V2-Flash	0.83	0.83	0.84
deepseek-ai/DeepSeek-V3.2	0.87	0.73	0.78
google/gemma-2-9b-it	0.70	0.66	0.79
google/gemma-3-12b-it	0.72	0.68	0.65
google/gemma-3-27b-it	0.78	0.80	0.78
google/gemma-4-26B-A4B-it	0.77	0.80	0.76
google/gemma-4-31B-it	0.68	0.66	0.59
google/gemma-4-E2B-it	0.76	0.80	0.82
google/gemma-4-E4B-it	0.82	0.71	0.77
moonshotai/Kimi-K2.5	0.78	0.71	0.82
moonshotai/Kimi-K2.6	0.73	0.76	0.76
openai/gpt-oss-120b	0.72	0.72	0.76
openai/gpt-oss-20b	0.74	0.74	0.74
zai-org/GLM-4.5-Air-FP8	0.84	0.81	0.77
zai-org/GLM-4.7-FP8	0.93	0.92	0.93
zai-org/GLM-4.7-Flash	0.88	0.78	0.82
zai-org/GLM-5-FP8	0.90	0.81	0.89
zai-org/GLM-5.1-FP8	0.89	0.87	0.84
Median	0.82	0.79	0.79

A priori, this difference might not be expected to substantially affect results. However, Goldowsky-Dill et al. [2025] themselves note that threshold selection is highly unstable in practice, with small changes (e.g., layer choice) producing large variations in recall. Consistent with this, once the calibration is corrected, TPR on the Roleplaying dataset falls from 96% to 73.2%.

F Targeted Apollo Detector

The Apollo probe [Goldowsky-Dill et al., 2025] described in Section 5 is trained on contrastive activations derived from the Facts-true-false dataset [Azaria and Mitchell, 2023], paired with honesty- and deception-inducing system prompts. We also experimented with a variant — which we refer to as *Targeted Apollo* — trained on the more varied contrastive datasets of Natarajan et al. [2026], with the aim of improving generalisation across the broader range of lie-inducing motivations in Varied Deception. All other aspects of the method (single-layer residual-stream probe, run-time mean aggregation over non-reasoning assistant tokens, and threshold calibration to a 1% positive rate on Alpaca completions [Taori et al., 2023]) are unchanged.

Table 4 reports balanced accuracy on Varied Deception for both the original Apollo probe and the Targeted Apollo variant across all 31 subject models. The median balanced accuracy is essentially unchanged (0.84 for Apollo vs. 0.81 for Targeted Apollo), though the per-model picture is inconsistent: the targeted training data improves performance on some subject models (e.g. Qwen 3.6 27B, 0.84 $\rightarrow$ 0.92; DeepSeek-V3.2, 0.85 $\rightarrow$ 0.91) while substantially degrading it on others (e.g. Qwen 3.5 27B, 0.95 $\rightarrow$ 0.82; Gemma 4 26B-A4B, 0.91 $\rightarrow$ 0.82).

Table 4: Balanced accuracy on Varied Deception for the original Apollo probe [Goldowsky-Dill et al., 2025] and the Targeted Apollo variant trained on the contrastive datasets of Natarajan et al. [2026].

model	Apollo	Targeted Apollo
MiniMaxAI/MiniMax-M2	0.72	0.66
MiniMaxAI/MiniMax-M2.7	0.83	0.80
Qwen/Qwen3-14B	0.87	0.84
Qwen/Qwen3-30B-A3B	0.86	0.80
Qwen/Qwen3-32B	0.74	0.75
Qwen/Qwen3-4B	0.64	0.61
Qwen/Qwen3-8B	0.74	0.75
Qwen/Qwen3.5-122B-A10B-FP8	0.86	0.88
Qwen/Qwen3.5-27B	0.95	0.82
Qwen/Qwen3.5-35B-A3B-FP8	0.88	0.91
Qwen/Qwen3.5-4B	0.89	0.89
Qwen/Qwen3.6-27B	0.84	0.92
Qwen/Qwen3.6-35B-A3B	0.87	0.94
XiaomiMiMo/MiMo-V2-Flash	0.83	0.80
deepseek-ai/DeepSeek-V3.2	0.85	0.91
google/gemma-2-9b-it	0.77	0.73
google/gemma-3-12b-it	0.74	0.78
google/gemma-3-27b-it	0.81	0.81
google/gemma-4-26B-A4B-it	0.91	0.82
google/gemma-4-31B-it	0.95	0.90
google/gemma-4-E2B-it	0.79	0.74
google/gemma-4-E4B-it	0.83	0.81
meta-llama/Llama-3.3-70B-Instruct	0.84	0.87
moonshotai/Kimi-K2.5	0.89	0.76
moonshotai/Kimi-K2.6	0.84	0.78
openai/gpt-oss-120b	0.80	0.79
openai/gpt-oss-20b	0.75	0.75
zai-org/GLM-4.5-Air-FP8	0.90	0.89
zai-org/GLM-4.7-FP8	0.95	0.89
zai-org/GLM-4.7-Flash	0.85	0.88
zai-org/GLM-5-FP8	0.94	0.89
zai-org/GLM-5.1-FP8	0.91	0.90
Median	0.84	0.81

G Unrelated Questions Activation Variant

The Unrelated Questions classifier described in Section 5 follows Pacchiardi et al. [2023] in training a logistic classifier on a 50-dimensional vector of yes-vs-no logprob differences over a fixed set of unrelated follow-up questions. We also experimented with an activation-based variant — which we refer to as *Unrelated Qs (Activations)* — that replaces the logprob features with residual-stream activations elicited by the same follow-up questions. Concretely, rather than forming the feature vector from yes-vs-no logprob differences, we extract residual-stream activations at a single layer (the 80th-percentile layer by depth) at the answer token of each follow-up question, and train an MLP classifier on these activations. All other aspects of the method — the set of 50 unrelated follow-up questions, rollout construction and filtering, and threshold calibration to a 1% positive rate on Alpaca completions [Taori et al., 2023] — are unchanged.

Table 5 reports balanced accuracy on Varied Deception for both variants across our subject models. The activation-based variant performs worse on aggregate, with median balanced accuracy falling from 0.81 (logprobs) to 0.76 (activations), and underperforming the logprob-based classifier on the large majority of subject models.

Table 5: Balanced accuracy on Varied Deception for the original Unrelated Questions classifier [Pacchiardi et al., 2023], trained on yes-vs-no logprob differences, and an activation-based variant trained on residual-stream activations elicited by the same follow-up questions.

model	Unrelated Qs	Unrelated Qs Activations
MiniMaxAI/MiniMax-M2.7	0.78	0.76
Qwen/Qwen3-14B	0.76	0.73
Qwen/Qwen3-32B	0.78	0.67
Qwen/Qwen3.5-122B-A10B-FP8	0.89	0.62
Qwen/Qwen3.5-27B	0.76	0.62
Qwen/Qwen3.5-35B-A3B-FP8	0.87	0.83
Qwen/Qwen3.5-4B	0.81	0.71
Qwen/Qwen3.6-27B	0.80	0.68
Qwen/Qwen3.6-35B-A3B	0.89	0.76
XiaomiMiMo/MiMo-V2-Flash	0.81	0.82
google/gemma-2-9b-it	0.77	0.69
google/gemma-3-27b-it	0.78	0.77
google/gemma-4-26B-A4B-it	0.93	0.82
google/gemma-4-31B-it	0.94	0.81
google/gemma-4-E2B-it	0.81	0.78
google/gemma-4-E4B-it	0.78	0.80
meta-llama/Llama-3.3-70B-Instruct	0.84	0.77
zai-org/GLM-4.5-Air-FP8	0.77	0.79
zai-org/GLM-4.7-FP8	0.88	0.73
zai-org/GLM-4.7-Flash	0.84	0.73
Median	0.81	0.76

H Harm Pressure Verification

The Liars’ Bench Harm Pressure Choice setting [Kretschmar et al., 2025] constructs deception samples by placing models under pressure to answer WMDP [Li et al., 2024] multiple-choice questions. The underlying hypothesis is that if a model answers correctly under multiple benign phrasings of a question but incorrectly under a harmful framing, it may be lying to avoid facilitating harm. Wang et al. [2025] use an independently rolled-out version of this dataset for their analysis of Sonnet 3.7.

We first used a helpful-only model to verify the WMDP dataset itself. This model flagged 20% of questions as having incorrect labelled answers, 8% as having multiple correct or ambiguous answers, and 2% as having no correct answer. While the helpful-only model may itself error on some questions, manual spot-checks revealed a comparably high error rate, suggesting that roughly one third of the WMDP dataset may be mislabelled.

We next conducted automated transcript analysis of the Harm Pressure Choice dataset [Kretschmar et al., 2025] across all Llama 3.3 70B samples, where we asked a helpful-only model if it agreed with the answer marked as canonically correct in each sample. Our transcript analyser disagreed with 13% of labels for samples marked as honest, and with 34% of labels for samples marked as lying. We are uncertain as to the impact this may have on downstream results.

Finally, we audited the labels from Wang et al. [2025]: only one label was flagged as outright incorrect, and we disagreed with the automated verifier in this case, though 15 questions were flagged as potentially ambiguous (i.e., having multiple plausibly correct answers).

I Llama 3.3 70B Introspection Challenges

In this appendix we present evidence that Llama-3.3-70B-Instruct exhibits unreliable introspection, both as the off-the-shelf instruct model and after fine-tuning into a model organism with a known hidden objective. We draw two conclusions: (i) simple self-report baselines are often not possible to construct without an unacceptably high error rate for real-world auditing use, and (ii) any technique that uses self-report labels — including auditing with lie-detection probes — may be unreliable on Llama 70B even if the technique has promise, as the model organism may not be able to introspect